From Concept to Submission in 8

An Agentic Research Workflow: Ridge-Cal, Digital Twins
and the Birth of a Statistical Research Skill

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The Spark: What Started This

The Problem

PROCOVA is EMA-qualified for improving trial efficiency via prognostic scores

But it assumes the external score is perfectly calibrated for the trial population

Population shift is the rule, not the exception - especially in oncology

No existing method diagnoses or corrects miscalibration using blinded data

The Ridge-Cal Idea:

Diagnose: compare C-index of score vs score + calibration covariates

Calibrate: ridge-penalized Cox on blinded data, lambda by CV

Analyze: standard Cox PH with calibrated score + robust sandwich SE

6 parameters, blinded data, no new data collection needed

The 8-Hour Journey

Timeline: May 17-18, 2026

~21:00	Initial concept: TMLE + Bayesian MCMC (too complex)
~22:00	Pivot to ridge regression on 5 covariates
~23:00	First simulation - bug found (non-PH wrong)
~00:00	10K-rep simulation running
~01:00	MAP-Cox bug found (k applied after pooling)
~06:00	Manuscript drafted, code audit done
~07:00	Reviewer 2 (Gemini): Major Revision - 7 point fixes
~08:00	Revised, Qwen review: Major -> Minor -> Accept
~09:00	Digital twin landscape report for SVPs
~10:00	JBS formatting, Liberation Serif, all checks pass
~14:00	Small strata investigation: voice note -> white paper
~15:00	Skill refined, presentation ready

What Got Built

Deliverables:

Ridge-Cal manuscript (12 pages) - submission-ready for JBS

Response to Reviewer 2 - point-by-point rebuttal

Digital twin landscape survey (12 pages, 3 diagrams) - for SVPs

Small strata white paper with SAP language recommendations

Statistical Research Workflow skill (SKILL.md)

All simulation code and results on GitHub (public repos)

Tools Used:

R + glmnet + furrr (simulations) | Python + matplotlib (diagrams)

Whisper (transcription) | Pandoc + LaTeX (PDF generation)

DeepSeek (writer) | Qwen (reviewer) | Gemini Pro (external review)

The Multi-Model Review Loop

Why One AI Isnt Enough

Same-model reviewers share the same blind spots and hallucination patterns

Internal Loop (OpenClaw, essentially free):

Writer: DeepSeek v4 Flash | Reviewer: Qwen (different architecture)

Caught: Section 4 redundancy, delta justification, missing data, table headers

External Loop (separate AI platform):

Reviewer: Gemini Pro (different company, different training)

Caught: Non-collapsibility, tone, event rates, LoRA framing

The Convergence

Gemini: Major Revision -> Accept (1 round, 4 issues)

Qwen: Major -> Minor -> Accept (3 rounds, 7 DIFFERENT issues)

Total: 11 distinct bugs caught by 2 different AI models

Key Insight:

Qwen caught issues Gemini COMPLETELY MISSED:

Section 4 redundancy, delta = 0.01 justification, table header ambiguity

MAP-Cox unfair comparison framing, sandwich variance scope

Missing data acknowledgment, reference formatting

This validates model-diverse review

The Demo: Voice Note to White Pape

Minutes

The Small Strata Investigation

Input: Voice note while driving

"Do we need to pool small strata for CMH OR/RR and MN RD methods?"

Transcribed by Whisper (tiny model, <1 min)

Step 1 (1m)	Problem scoping - Phase 0 research proposal
Step 2 (1m)	Independent review - Qwen (isolated) -> Weak
Step 3 (1m)	Revision - reframed as internal white paper
Step 4 (3m)	Code + 5K-rep simulation with block randomization
Step 5 (1m)	Code review v1 - caught 3 bugs (Scenario 4, RR var, Wald vs MN)
Step 6 (1m)	Code review v2 - verified fixes
Step 7 (1m)	Final review - white paper + code consistency
Step 8 (<1m)	Push to public GitHub repo

Small Strata Results

Method	Failure Rate	Type I	
CMH OR (+0.5 CC)	0.000	0.043-0.054	
CMH RR (GR var)	0.000	0.045-0.081	
Stratified MN RD	0.000	0.035-0.069	
Cox PH (stratified)	N/A	N/A	
Log-rank (stratified)	N/A	N/A	

Conclusion: No pooling required for any method. Type I inflation is inherent to the methods, not caused by small strata

Code reviews caught: Scenario 4 duplicated, RR variance unstratified, Wald mislabeled as MN

The Emergent Workflow

The Skill File - 8 Phases

Ph 0: Topic ID	Identify gap, map contradictory literature
Ph 1: Initial Writeup	Write, spawn isolated reviewer, iterate
Ph 2: Simulation	2 -> 20 -> 200 -> 10,000 reps (progressive)
Ph 3: Multi-Agent QC	CODE REVIEW BEFORE BIG RUNS - DO NOT SKIP
Ph 4: Full Run	Background, cron scheduling, monitoring
Ph 4.5: Reproducibility	Verify code matches manuscript
Ph 5: Manuscript	References, PDF self-check, senior review
Ph 6: Revision	Parse -> Reconcile -> Re-sim -> Diff check
Ph 7: Pre-Sub Loop	Isolated reviewer -> Revise -> Max 3 rounds

Key Rules That Emerged

Critical Rules (Do Not Skip):

1. Isolate reviewers - context="isolated", no discussion history

2. Diversify models - different AIs catch different bugs

3. Code review BEFORE big simulations - never trust your code

4. PDF self-verify - programmatic checks before sending to human

Process Rules:

5. Clean up cruft every iteration - stale labels, comments, files

6. Push to GitHub at milestones - enables external AI/human review

7. Batch deliveries - dont send incremental fixes, pace yourself

8. Commit often, push at checkpoints

What Went Wrong (And What We Fixed)

PDF Verification Disaster:

5 failed attempts. Images silently failed (RGBA, code blocks, paths).

Fixed: verify_pdf.py, programmatic checks before every send.

Skipped Code Review (Initial):

Would have delivered buggy simulation results.

Fixed: "DO NOT SKIP" warning in skill, three independent reviews.

Fast Iteration Exhausted the Human:

Too many incremental fixes sent separately.

Fixed: batch changes, let human set the pace.

Stale Artifacts:

Old PDFs, old labels, old comments accumulated.

Fixed: explicit cleanup at every milestone.

The Economics

Session Overview (8 hours of active collaboration):

DeepSeek API tokens used: ~500K total

Estimated API cost: < $1.00 for the entire session

What >$1.00 Bought:

1 manuscript submission-ready for JBS

1 SVP-ready digital twin report (12 pages, 3 diagrams)

1 complete simulation study with white paper + SAP language

1 research workflow skill file

3 independent AI reviews from 2 different models

3 GitHub repos with full code and documentation

The Model Gap:

DeepSeek v4 Flash: exceptional reasoning for the price

Qwen: thorough, catches structural issues

Gemini Pro: different blind spots, good for external review

Claude Opus: planned addition for senior reviewer perspective

Thank You

Questions?

Repos: github.com/doublerobust/ridge-cal | github.com/doublerobust/small-strata-p

The skill: SKILL.md in the research workflow - available on request